0.1 Quotient group

Let $(G,+)$ be an abelian group and let H be a subgroup of G. Consider the binary relation $\mathcal{R}$ defined on G by:

$$\forall x, y \in G,\ x\mathcal{R}y \text{ if and only if } x - y \in H.$$

This relation $\mathcal{R}$ is an equivalence relation. We denote the quotient set $G/\mathcal{R}$ by G/H.

We define a binary operation on G/H as follows:

$$\forall\ \overline{x}, \overline{y} \in G/H:\ \ \overline{x} \oplus \overline{y} = \overline{x+y}.$$

Then $(G, \oplus)$ is a group called *quotient group*. Indeed, the associativity of $\oplus$ follows from the associativity of $+$ in G. The neutral element is $\overline{e}$, where e is the neutral element of G, and the symmetric element of an element $\overline{x}$ is $\overline{-x}$, where $-x$ is the symmetric of x in G.

Example 1 *For any nonnegative integer n, $n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$. Let $\mathcal{R}$ be the equivalence relation defined on $\mathbb{Z}$ by:*

$$x\mathcal{R}y \textit{ if and only if } x - y \in n\mathbb{Z}.$$

The equivalence class of $x \in \mathbb{Z}$ is denoted by $\overline{x}$.

The quotient set of $\mathbb{Z}$ by $\mathcal{R}$ is:

$$\mathbb{Z}/n\mathbb{Z} = \{\overline{x} : x \in \mathbb{Z}\} = \{r : r \in \{0, 1, \ldots, n-1\}\},$$

.

Let G and G' be two groups, and f be a homomorphism from G to G'. We assume that G is commutative. We consider in G the binary relation defined by $x\mathcal{R}y \Leftrightarrow f(x) = f(y)$. The equivalence classes are thus the classes of G modulo $\ker f$.

The canonical decomposition of f is then given as follows:

$$\begin{array}{ccc} G & \xrightarrow{f} & S \\ \downarrow s & & \uparrow i \\ G/\ker f & \xrightarrow{h} & f(G) \end{array}$$

where s is the canonical surjection from G into $G/\ker f$, the map i is the canonical injection map from $f(G)$ into G', and h is defined by $h(\overline{x}) = f(x)$. The map h is an isomorphim of groups.

0.1.1 Cyclic groups

Definition 2 *A group G is said to be cyclic (or monogenous) if there exists an element $a \in G$ such that for every $x \in G$, there exists an integer $n \in \mathbb{Z}$ with $x = a^n$. In other words, $G = \{a^n : n \in \mathbb{Z}\}$, where the element a is called a generator of G, and we write $G = \langle a \rangle$.*

- ▶ When the binary operation of G is denoted additively, and G is a cyclic group generated by a, then $G = \{na : n \in \mathbb{Z}\}$.

Example 3

1. The group $(\mathbb{Z}, +)$ is a cyclic. We have $\mathbb{Z} = \langle +1 \rangle = \langle -1 \rangle$.
2. The group $(\mathbb{Z}/n\mathbb{Z}, \oplus)$ is a cyclic. We have $\mathbb{Z}/n\mathbb{Z} = \langle \overline{1} \rangle$.
3. Let $n \in \mathbb{N}$ and $U_n = \{z \in \mathbb{C} : z^n = 1\}$. The group$(U_n, \times)$ is a cyclic, and $U_n = \langle \zeta_1 \rangle$, where $\zeta_1 = e^{(2\pi i/n)}$.

Let G be a group and let a be an element of G. The set $H = \{a^n : n \in \mathbb{Z}\}$ is a subgroup of G. This subgroup is commutative even though G is not. We call it the subgroup generated by a and we write $H = \langle a \rangle$. Two cases arise:

- If for any $n \in \mathbb{Z}$, we have $a^n \neq e$, we say that a is of *infinite order*. In this case, H is infinite and the map defined by

$$\begin{array}{ccc} \mathbb{Z} & \to & H \\ n & \mapsto & a^n \end{array}$$

  is a group isomorphism.

- If there exists $n \in \mathbb{Z}$ such that $a^n = e$. The least positive integer m satisfying this property is called the *order* of a, and H is necessarily finite,

$$H = \{e, a, a^2, \ldots, a^{m-1}\},$$

  and we have $a^\ell = a^{\ell'}$ if and only if $\ell - \ell' \in m\mathbb{Z}$. In this case, the map defined by

$$\begin{array}{ccc} \mathbb{Z}/m\mathbb{Z} & \to & H \\ \overline{k} & \mapsto & a^k \end{array}$$

  is a group isomorphism, where $\mathbb{Z}/m\mathbb{Z}$ is endowed with its canonical group structure.

We conclude that a cyclic group is either isomrphic to $\mathbb{Z}$ (when G is infinite) or to $\mathbb{Z}/m\mathbb{Z}$ for some positive integer m (when G is finite).

Here are some notes on cyclic groups:

Proposition 4

1. *If a cyclic group is generated by a, then it is also generated by a^{-1}.*
2. *Every cyclic group is abelian.*
3. *Every subgroup of a cyclic group is also cyclic.*
4. *If an infinite cyclic group G is generated by a, then a and a^{-1} are the only generators of G.*

Proof.

1. Let $G = \langle a \rangle$. Since $a = (a^{-1})^{-1}$, all the elements obtained as powers of a can also be obtained as powers of a^{-1}.

2. Let G be a cyclic group. Then $G = \langle a \rangle$ for some $a \in G$. For any $x, y \in G$ we have $x = a^n$ and $y = a^m$ for some $n, m \in \mathbb{Z}$. Therefore

$$xy = a^n a^m = a^{n+m} = a^{m+n} = a^m a^n = yx,$$

so commutativity holds.

3. Let G be a cyclic group. Then $G = \langle a \rangle$ for some $a \in G$. Let H be a subgroup of G. Every $h \in H$ can be written as $h = a^n$ for some $n \in \mathbb{Z}$. Let m be the smallest positive integer such that $a^m \in H$. Define $h_0 = a^m$. Then $H = \langle h_0 \rangle$. Indeed, $h_0 \in H$, then all the powers of h_0 are in H. Hence $\langle h_0 \rangle \subset H$. Conversely, let $t \in H$. Then $t = a^\ell$ for some $\ell \in \mathbb{Z}$. The Euclidian division of ℓ by m gives $\ell = mk + r$ with $0 \leq r < m$. We have $\mathrm{a}^\ell \in H$ and $a^{mk} \in H$ then $a^{\ell - mk} = a^r \in H$. Since m is the smallest integer satisfying the property, necessarily $r = 0$, hence $t = h_0^k$. Thus, $t \in \langle h_0 \rangle$ and we have $H \subset \langle h_0 \rangle$.

4. Let G be an infinite cyclic group such that $G = \langle a \rangle$ for some $a \in G$. Suppose that G is generated by a^k for some $k \in \mathbb{Z}$. Then there exists $k' \in \mathbb{Z}$ such that $(a^k)^{k'} = a$. This gives $a^{kk'-1} = e$, which implies that $kk' - 1 = 0$ (since a is of infinite order). Then $kk' = 1$. Hence $k = k' = 1$ or $k = k' = -1$. Consequently, the only generators of G are a and a^{-1}.

■

0.1.2 Finite groups

Recall that a group G with a finite number of elements is called a *finite group*. This number is called the order of the group and is denoted by $|G|$.

Theorem 5 (Lagrange) *Let G be a finite group. Then the order of any subgroup of G divides the order of G.*

Proof. Let H be a subgroup of the finite group G. Consider the binary relation on G defined by:

$$\forall x, y \in G : x\mathcal{R}y \text{ if and only if } x^{-1}y \in H.$$

We can easily show that $\mathcal{R}$ is an equivalence relation. Since G is of finite order, there are a finite number of classes. On the one hand, for any $a \in G$, the class of a, denoted by aH, is given by:

$$aH = \{ah, h \in H\}.$$

On the other hand the map

$$\begin{array}{ccc} H & \longrightarrow & aH \\ t & \longmapsto & at \end{array}$$

is bijective. Therefore $|aH| = |H|$. Since the classes aH $(a \in G)$ have the same cardinality as H and they form a partition of G, then the order of G is the number of classes multiplied by the order of $|H|$. This shows that the order of H divides the order of G. ■

Corollary 6 *If G is a group of finite order n, then the order of any element of G divides the order of G.*

Proof. Let ℓ be the order of a an element of G. Then ℓ is the least positive integer, such that $a^\ell = e$. Then, we can show that the set $H = \left\{a, a^2, a^3, \ldots, a^{\ell-1}, a^\ell = e\right\}$ is a subgroup of G. Since this subgroup has order ℓ, thus, ℓ divides the order of G. ■

Corollary 7 *Let G be a finite group of prime order p. Then*

1. *For any $x \in G \setminus \{e\}$ we have $G = \langle x \rangle$.*

2. *The only subgroups of G are $\{e\}$ and G.*

Proof.

1. It is clear that if $x \in G \setminus \{e\}$, then the order of $\langle x \rangle$ is different from 1 and divides $|G| = p$. Since p is prime, $|\langle x \rangle| = |G|$, necessarily $\langle x \rangle = G$. In other words, G is cyclic generated by any element x belonging to G with $x \neq e$.

2. Since the order of a subgroup divides the order the group and $|G| = p$ is prime, the only subgroups of G are of order 1 and p which corresponds to $\{e\}$ and G, respectively.

■

Theorem 8 *Let $G =< a >$ be a finite cyclic group of order n and let k be a positive integer. Then $G =< a^k >$ if and only if $k < n$ and $\gcd(k, n) = 1$.*

Proof. Since $G = \langle a \rangle$, we have $G = \{e, a, a^2, \ldots, a^{n-1}\}$. Let $k \in \mathbb{N}$ and put $H = \langle a^k \rangle$. It is clear that $H = G$ if and only if $a \in H$ (since then H contains all powers of a and therefore all elements of G). We have $a \in H$ if and only if there exists $u \in \mathbb{Z}$ such that $a = a^{ku}$. This is equivalent to $a^{ku-1} = e$. Thus, there exists $u \in \mathbb{Z}$ such that $ku - 1 = vn$ for some $v \in \mathbb{Z}$, which is equivalent to the existence of $u, v \in \mathbb{Z}$ such that $ku + nv = 1$. This last condition is equivalent to $\gcd(k, n) = 1$. ■

0.1.3 A remarkable finite group:

Symmetric group: Let n be a positive integer and $E = \{1, 2, ..., n\}$. The group $S(E)$ of bijections from E into E equipped with the composition of maps, is a group of order $n!$, which we call the *symmetric group* on n elements and denote simply S_n. It is also called the group of permutations of S. Its $n!$ elements can be described as follows: each element σ of S_n can be written as a table (matrix) with 2 rows and n columns. The first row contains the elements $1, 2, ..., n$ and the second one indicates the respective images by σ of $1, 2, ..., n$ in the first row. For instance, the symmetric group S_3 is of order $3! = 6$. We then have $id_S = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \tau_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \tau_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \tau_3 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \sigma_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$.

$$S_3 = \{id_E, \tau_1, \tau_2, \tau_3, \sigma_1, \sigma_2\}.$$

$\circ$	id_S	τ_1	τ_2	τ_3	σ_1	σ_2
id_S	id_S	τ_1	τ_2	τ_3	σ_1	σ_2
τ_1	τ_1	id_S	σ_1	σ_2	τ_2	τ_3
τ_2	τ_2	σ_2	id_S	σ_1	τ_3	τ_1
τ_3	τ_3	σ_1	σ_2	id_S	τ_1	τ_2
σ_1	σ_1	τ_3	τ_1	τ_2	σ_2	id_S
σ_2	σ_2	τ_2	τ_3	τ_1	id_S	σ_1

Table of multiplication of S_3

In particular, we can deduce that the group S_3 is not abelian since $\tau_1 \circ \tau_2 \neq \tau_2 \circ \tau_1$. We can also see that the group S_3 admits three subgroups of order 2 which are $H_1 = \{id_S, \tau_1\}$, $H_2 = \{id_S, \tau_2\}$ and $H_3 = \{id_S, \tau_3\}$, and one subgroup of order 3 which is $H_4 = \{id_S, \sigma_1, \sigma_2\}$. This is an example also of a non-abelian group containing abelian subgroups.